

FRAME1: Tract-Level Sampling Frame

Detroit Transportation Survey Sample Design

Team Effect Size (Kevin Linares, Tavril Saint Jean)

2026-04-17

This file documents the cleaned and Kish-linked tract-level sampling frame used for the Detroit Transportation Survey. Each row represents one primary sampling unit (PSU). The `n_tracts` column indicates how many original Census tracts were combined into each receiving unit (1 means not combined, greater than 1 means the receiving tract absorbed one or more undersized neighbors). The composite measure of size (MOS) and PSU inclusion probability π_i were used to select 40 tracts via systematic probability-proportional-to-size sampling with `set.seed(-77)`.

The frame contains 265 PSUs covering 480,131 adults aged 18 and older. Source data are from the 2020 Decennial Census Demographic and Housing Characteristics Summary File, retrieved through the U.S. Census Bureau API.

Table 1

Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163500100	1	1,492	2,752	1,087	701	672	292	12.3832	0.229909
26163500200	1	1,076	2,088	749	554	496	289	9.4056	0.174626
26163500300	1	917	1,464	598	375	354	137	6.5900	0.122351
26163500400	1	406	551	143	166	146	96	2.4879	0.046191
26163500500	1	644	1,076	402	236	268	170	4.8201	0.089491
26163500600	1	1,178	1,860	706	454	474	226	8.3552	0.155124
26163500700	1	1,679	2,779	1,008	778	702	291	12.5341	0.232711
26163500800	1	1,009	1,784	636	483	471	194	8.0345	0.149171
26163500900	1	1,540	2,573	940	679	647	307	11.5842	0.215074
26163501000	1	1,163	2,029	752	497	521	259	9.1139	0.169211
26163501100	1	1,345	2,287	850	573	514	350	10.2897	0.191040
26163501200	1	766	1,102	393	273	296	140	4.9491	0.091887
26163501300	1	1,308	2,175	802	536	491	346	9.7806	0.181587
26163501400	1	1,505	2,685	867	686	684	448	12.0706	0.224104
26163501500	1	1,578	2,893	1,027	823	701	342	13.0583	0.242442
26163501600	1	907	1,467	484	338	413	232	6.5708	0.121995
26163501700	1	1,045	1,765	481	401	465	418	7.9027	0.146722

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163501800	1	1,896	3,202	979	759	852	612	14.3566	0.266547
26163501900	1	1,112	1,720	591	475	441	213	7.7520	0.143925
26163502000	1	766	1,289	437	357	346	149	5.8076	0.107826
26163502600	1	1,546	2,318	747	563	514	494	10.4159	0.193384
26163502700	1	1,278	1,852	506	429	560	357	8.2866	0.153851
26163503100	1	1,122	2,221	708	506	611	396	9.9460	0.184659
26163503200	1	1,202	2,068	783	572	498	215	9.3285	0.173194
26163503300	1	1,360	2,308	805	644	547	312	10.4123	0.193316
26163503400	1	494	673	203	178	190	102	3.0255	0.056172
26163503500	1	1,239	1,937	681	510	473	273	8.7210	0.161915
26163503600	1	643	1,080	377	273	238	192	4.8600	0.090232
26163503900	1	826	1,265	433	287	298	247	5.6724	0.105315
26163504000	1	469	772	249	215	168	140	3.4838	0.064680
26163504100	1	611	895	266	208	250	171	4.0087	0.074427
26163504200	1	875	1,218	390	293	312	223	5.4657	0.101478
26163504300	1	912	1,402	428	342	353	279	6.2938	0.116852
26163504400	1	715	1,045	313	283	278	171	4.7035	0.087326
26163505100	1	1,441	2,253	692	495	637	429	10.0763	0.187077
26163505200	1	1,340	2,204	731	506	548	419	9.8811	0.183455
26163505400	1	900	1,417	459	352	351	255	6.3664	0.118200
26163505600	1	1,794	2,992	980	706	806	500	13.4148	0.249062
26163505700	1	1,635	2,299	611	540	631	517	10.2987	0.191207
26163505800	1	1,199	2,221	857	555	470	339	9.9974	0.185613
26163506100	1	744	1,379	507	319	336	217	6.1868	0.114865
26163506200	1	778	1,353	447	323	333	250	6.0727	0.112747
26163506300	1	826	1,361	444	332	344	241	6.1108	0.113455
26163506400	1	342	697	206	179	209	103	3.1287	0.058088
26163506500	1	932	1,459	423	362	408	266	6.5463	0.121540
26163506600	1	1,257	1,970	629	468	529	344	8.8338	0.164010
26163506700	1	457	770	207	184	223	156	3.4494	0.064043
26163506800	1	1,637	2,906	924	701	748	533	13.0408	0.242117
26163506900	1	1,459	2,551	850	633	579	489	11.4693	0.212940
26163507000	1	1,183	1,886	634	467	449	336	8.4763	0.157372
26163507100	1	1,060	1,690	513	389	472	316	7.5681	0.140510
26163507200	1	519	750	231	188	185	146	3.3701	0.062570
26163507300	1	1,086	1,687	517	362	438	370	7.5457	0.140095
26163507400	1	1,206	1,891	590	438	519	344	8.4715	0.157284
26163507500	1	1,072	1,681	474	377	419	411	7.5280	0.139766

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163508000	1	629	835	270	210	270	85	3.7442	0.069515
26163508100	1	679	963	277	252	259	175	4.3290	0.080372
26163509000	1	1,090	1,511	456	391	401	263	6.7918	0.126097
26163509100	1	1,440	3,214	1,293	771	778	372	14.4392	0.268080
26163510600	1	1,636	3,661	1,424	968	815	454	16.5019	0.306376
26163511200	1	736	940	351	227	199	163	4.2266	0.078471
26163511300	1	822	1,511	582	375	324	230	6.7995	0.126240
26163511400	1	1,303	1,675	508	330	436	401	7.4763	0.138807
26163511900	1	1,234	1,320	499	272	397	152	5.8949	0.109446
26163512800	1	1,357	1,819	529	427	564	299	8.1414	0.151154
26163513200	1	915	1,183	326	302	338	217	5.3106	0.098598
26163513300	1	1,080	1,669	404	354	455	456	7.4554	0.138419
26163513700	1	1,479	2,307	580	457	609	661	10.2922	0.191087
26163513800	1	1,636	1,945	718	360	534	333	8.6711	0.160989
26163513900	1	905	1,165	380	238	325	222	5.2025	0.096591
26163514100	1	1,021	1,304	389	285	373	257	5.8301	0.108243
26163514200	1	1,108	1,485	368	286	423	408	6.6163	0.122839
26163514300	1	1,040	1,726	390	331	518	487	7.6840	0.142663
26163514500	1	467	654	206	134	186	128	2.9200	0.054212
26163515200	1	664	1,120	258	205	316	341	4.9840	0.092535
26163515300	1	1,906	2,439	703	418	713	605	10.8415	0.201285
26163515400	1	725	1,128	330	302	237	259	5.0836	0.094384
26163515700	1	3,906	4,417	1,048	578	1,168	1,623	19.5478	0.362928
26163516000	1	1,080	1,722	537	386	465	334	7.7088	0.143122
26163516400	1	739	978	351	238	255	134	4.3912	0.081527
26163516500	1	1,928	2,441	965	504	522	450	10.9327	0.202978
26163516600	1	1,436	1,679	638	319	377	345	7.5019	0.139281
26163516700	1	1,881	2,604	699	502	712	691	11.6085	0.215526
26163516800	1	405	474	121	97	164	92	2.1110	0.039193
26163516900	1	1,584	1,948	256	221	608	863	8.5817	0.159329
26163517000	1	2,430	2,852	1,280	485	474	613	12.7446	0.236619
26163517100	1	1,155	1,479	541	223	324	391	6.5782	0.122133
26163517200	1	941	1,650	672	417	417	144	7.4217	0.137793
26163517300	1	1,333	1,683	699	428	302	254	7.5891	0.140901
26163517500	1	2,126	2,528	780	344	655	749	11.2044	0.208023
26163518000	1	1,526	2,016	878	387	401	350	9.0216	0.167497
26163518900	1	1,016	1,462	462	302	340	358	6.5393	0.121409
26163519000	1	636	879	252	172	270	185	3.9165	0.072714

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163519100	1	712	861	210	173	263	215	3.8372	0.071243
26163519200	1	832	1,164	276	290	350	248	5.2173	0.096865
26163519300	1	699	1,057	254	195	327	281	4.7010	0.087279
26163520200	1	1,063	3,550	3,241	137	96	76	15.7598	0.292598
26163520300	1	1,676	2,505	1,809	312	231	153	11.1897	0.207750
26163520400	1	1,379	1,583	906	333	228	116	7.1204	0.132199
26163520700	1	3,303	3,067	1,776	466	412	413	13.7072	0.254489
26163520800	1	1,079	1,185	399	233	293	260	5.2928	0.098267
26163521100	1	841	1,259	433	334	281	211	5.6729	0.105325
26163521400	1	978	1,211	604	289	192	126	5.4587	0.101347
26163521500	1	705	1,031	413	224	254	140	4.6190	0.085758
26163521800	1	901	1,179	282	207	249	441	5.2540	0.097546
26163521900	1	1,530	2,409	1,125	436	488	360	10.7686	0.199931
26163522000	1	915	1,341	463	300	372	206	6.0034	0.111461
26163522300	1	1,005	982	257	192	368	165	4.3651	0.081044
26163522400	1	519	614	262	141	168	43	2.7532	0.051115
26163522500	1	1,354	1,181	263	247	383	288	5.2643	0.097738
26163522800	1	648	913	293	242	248	130	4.1068	0.076247
26163523100	1	575	948	411	266	194	77	4.2852	0.079561
26163523200	1	1,157	2,252	880	658	465	249	10.1886	0.189163
26163523300	1	1,057	2,023	801	589	421	212	9.1515	0.169907
26163523400	1	1,260	1,669	627	447	373	222	7.5248	0.139706
26163523800	1	1,270	2,544	1,063	729	522	230	11.5056	0.213616
26163524001	1	1,013	1,797	666	550	367	214	8.1421	0.151167
26163524101	1	1,519	2,583	897	714	607	365	11.6499	0.216294
26163524200	1	1,463	2,884	1,122	901	577	284	13.0806	0.242857
26163524300	1	880	1,776	672	555	382	167	8.0508	0.149472
26163524600	2	705	1,087	348	283	301	155	4.8860	0.090714
26163524700	1	1,858	2,954	743	645	878	688	13.1963	0.245005
26163524800	1	1,174	1,958	536	466	527	429	8.7764	0.162944
26163525400	1	844	1,312	413	288	395	216	5.8647	0.108886
26163525700	1	1,483	2,906	1,057	956	606	287	13.1989	0.245054
26163525800	1	640	1,140	460	352	233	95	5.1684	0.095958
26163525900	1	962	1,771	634	547	382	208	8.0231	0.148958
26163526000	1	965	1,948	834	538	383	193	8.8028	0.163434
26163526100	1	747	1,679	741	468	284	186	7.5967	0.141042
26163526200	1	875	1,869	810	580	300	179	8.4888	0.157604
26163526300	1	1,106	2,104	827	685	390	202	9.5619	0.177529

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163526400	1	508	964	363	265	205	131	4.3513	0.080787
26163526500	1	969	1,312	451	315	314	232	5.8916	0.109384
26163527900	1	1,452	1,386	421	368	357	240	6.2364	0.115786
26163530100	1	1,389	1,867	535	407	574	351	8.3402	0.154846
26163530200	1	1,583	2,903	1,364	482	627	430	12.9492	0.240417
26163530300	1	780	1,026	303	265	275	183	4.6108	0.085605
26163530400	1	782	901	329	184	245	143	4.0259	0.074746
26163530500	1	1,177	1,713	571	387	436	319	7.6753	0.142501
26163530800	1	1,011	1,092	333	231	310	218	4.8789	0.090583
26163530900	1	1,273	1,281	373	269	360	279	5.7220	0.106236
26163531100	1	774	785	231	159	282	113	3.4951	0.064890
26163531200	1	1,134	1,793	593	430	433	337	8.0498	0.149453
26163531300	1	667	893	253	190	277	173	3.9863	0.074010
26163531400	1	891	1,154	361	253	305	235	5.1640	0.095877
26163531501	1	860	982	325	259	244	154	4.4200	0.082063
26163531502	1	625	580	185	140	165	90	2.6007	0.048285
26163531600	1	1,015	1,290	394	294	387	215	5.7717	0.107158
26163531700	1	1,005	1,247	369	246	373	259	5.5591	0.103210
26163531800	1	960	1,127	349	255	322	201	5.0439	0.093645
26163531900	1	527	657	179	159	206	113	2.9422	0.054626
26163532100	1	1,160	1,780	590	425	412	353	7.9929	0.148398
26163532400	1	959	1,255	442	296	338	179	5.6281	0.104493
26163532600	1	1,490	1,733	627	337	454	315	7.7369	0.143644
26163532700	1	499	406	143	103	85	75	1.8279	0.033936
26163533000	1	794	1,131	322	223	348	238	5.0400	0.093573
26163533100	1	977	1,055	311	207	244	293	4.7124	0.087490
26163533200	1	703	839	292	164	209	174	3.7471	0.069569
26163533300	1	761	922	275	177	285	185	4.1065	0.076242
26163533400	1	1,468	1,646	551	360	429	306	7.3676	0.136789
26163533600	1	991	1,035	302	275	279	179	4.6549	0.086424
26163533800	1	1,184	1,445	410	345	420	270	6.4739	0.120195
26163533900	1	2,610	2,917	1,262	566	704	385	13.0393	0.242090
26163534100	1	1,127	1,316	359	321	404	232	5.8959	0.109464
26163534200	1	1,632	2,257	670	519	608	460	10.1093	0.187692
26163534300	1	676	820	283	194	207	136	3.6793	0.068311
26163534400	1	874	1,135	348	251	328	208	5.0760	0.094242
26163534700	1	1,463	2,214	669	521	594	430	9.9238	0.184246
26163534800	1	1,194	1,641	471	427	406	337	7.3804	0.137025

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163535000	1	1,241	1,808	561	449	464	334	8.1198	0.150753
26163535100	1	819	1,267	439	295	311	222	5.6838	0.105527
26163535200	1	1,057	1,516	517	387	344	268	6.8221	0.126660
26163535300	1	1,629	2,862	914	711	667	570	12.8639	0.238832
26163535600	1	1,613	2,896	816	614	740	726	12.9481	0.240396
26163535700	1	621	897	284	223	214	176	4.0310	0.074841
26163535800	1	1,864	2,966	890	702	814	560	13.2938	0.246814
26163536100	1	1,763	2,417	788	530	545	554	10.8302	0.201076
26163536200	1	878	1,545	487	376	384	298	6.9367	0.128787
26163536300	1	1,365	1,997	715	464	473	345	8.9615	0.166380
26163536400	1	1,242	1,609	462	382	446	319	7.2107	0.133876
26163536500	1	1,126	1,597	488	400	390	319	7.1763	0.133237
26163536600	1	1,404	1,905	530	472	529	374	8.5467	0.158679
26163536700	1	575	805	226	180	205	194	3.6041	0.066914
26163536800	1	735	1,154	398	301	254	201	5.1977	0.096501
26163536900	1	1,384	2,418	796	613	592	417	10.8717	0.201846
26163537000	1	1,497	2,381	810	549	541	481	10.6838	0.198357
26163537100	1	972	1,432	443	393	352	244	6.4527	0.119802
26163537200	1	350	450	122	131	126	71	2.0288	0.037667
26163537300	1	1,069	1,511	433	385	397	296	6.7882	0.126031
26163537500	1	1,170	1,988	643	477	473	395	8.9257	0.165716
26163537600	1	1,418	2,522	756	587	653	526	11.3039	0.209871
26163537700	1	1,420	2,165	710	512	555	388	9.7114	0.180304
26163537800	1	734	1,125	380	292	266	187	5.0634	0.094009
26163538100	1	1,020	1,803	425	393	481	504	8.0605	0.149652
26163538200	1	769	1,624	322	363	418	521	7.2644	0.134872
26163538300	1	1,738	1,772	595	371	544	262	7.9118	0.146891
26163538400	1	1,378	2,707	698	619	677	713	12.1258	0.225129
26163538500	1	2,401	3,943	1,260	900	944	839	17.6778	0.328208
26163538600	1	1,958	3,619	1,097	828	943	751	16.2131	0.301015
26163538700	1	1,706	2,911	815	650	678	768	13.0411	0.242123
26163538800	1	678	1,219	365	265	321	268	5.4532	0.101245
26163538900	1	1,618	2,786	783	576	682	745	12.4526	0.231198
26163539000	1	1,217	2,061	608	524	529	400	9.2610	0.171941
26163539100	1	804	1,362	391	298	351	322	6.0942	0.113147
26163539200	1	2,313	3,908	976	868	1,062	1,002	17.4792	0.324521
26163539300	1	1,586	2,945	788	700	771	686	13.2022	0.245114
26163539400	1	1,434	2,819	838	630	663	688	12.6304	0.234499

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Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163539500	1	1,355	2,379	519	516	563	781	10.6423	0.197586
26163539600	1	1,317	2,439	720	564	614	541	10.9319	0.202964
26163539700	1	1,020	1,675	424	338	449	464	7.4751	0.138783
26163540100	1	1,397	2,361	745	538	557	521	10.5854	0.196530
26163540200	1	1,476	2,322	826	524	588	384	10.4064	0.193206
26163540300	1	1,608	2,770	883	614	642	631	12.4121	0.230446
26163540400	1	1,252	2,026	588	458	543	437	9.0704	0.168402
26163540500	1	1,389	2,563	748	660	610	545	11.5273	0.214017
26163540600	1	1,371	2,416	720	584	560	552	10.8490	0.201424
26163540700	1	1,313	2,400	776	607	556	461	10.7938	0.200399
26163540800	1	920	1,627	464	414	389	360	7.3142	0.135798
26163540900	1	1,439	2,641	729	600	642	670	11.8327	0.219688
26163541000	1	1,732	2,729	867	635	641	586	12.2428	0.227302
26163541100	1	1,238	2,066	736	539	475	316	9.3037	0.172733
26163541200	1	1,124	1,563	535	356	441	231	6.9994	0.129952
26163541300	1	1,119	1,920	710	540	463	207	8.6647	0.160870
26163541400	1	1,185	2,025	603	448	582	392	9.0560	0.168136
26163541500	1	1,794	2,897	795	621	768	713	12.9513	0.240457
26163541700	1	1,524	2,054	635	485	489	445	9.2166	0.171118
26163541800	1	1,267	1,966	587	481	510	388	8.8238	0.163825
26163542100	1	1,957	3,419	985	884	804	746	15.3798	0.285545
26163542200	1	1,637	2,755	811	665	619	660	12.3730	0.229719
26163542300	1	1,046	1,740	578	435	429	298	7.8199	0.145186
26163542400	1	1,112	1,923	644	510	390	379	8.6694	0.160958
26163542500	1	1,032	1,972	571	442	493	466	8.8315	0.163967
26163542600	1	1,376	2,343	744	585	521	493	10.5359	0.195611
26163542900	1	1,362	2,682	717	609	683	673	12.0110	0.222997
26163543000	1	709	1,413	349	311	343	410	6.3239	0.117410
26163543100	1	966	1,915	487	427	489	512	8.5707	0.159125
26163543200	1	1,187	2,291	739	587	524	441	10.3084	0.191387
26163543300	1	1,412	2,516	753	600	592	571	11.2927	0.209662
26163543400	1	1,386	2,344	711	591	588	454	10.5327	0.195553
26163543500	1	542	822	319	212	183	108	3.7023	0.068738
26163543700	1	703	1,157	441	329	274	113	5.2245	0.096999
26163543900	1	487	601	219	132	193	57	2.6862	0.049872
26163544000	1	1,250	1,985	694	473	568	250	8.8996	0.165231
26163544100	1	1,059	1,620	630	430	350	210	7.3044	0.135615
26163544200	1	1,103	1,488	440	311	366	371	6.6534	0.123528

(continued)

Tract FIPS	n_tracts	TotHH	TotAdult	Age 18-34	Age 35-49	Age 50-64	Age 65+	MOS	pi_i
26163544300	1	1,320	1,795	561	490	496	248	8.0797	0.150010
26163544800	1	886	1,289	423	372	334	160	5.8165	0.107990
26163545100	1	530	753	246	196	181	130	3.3886	0.062914
26163545200	1	1,234	1,900	620	537	396	347	8.5808	0.159313
26163545500	1	1,543	2,590	993	716	576	305	11.6901	0.217040
26163545600	1	1,684	3,326	1,384	900	706	336	15.0112	0.278701
26163545700	1	1,106	1,987	751	584	428	224	8.9879	0.166871
26163545800	1	1,859	3,508	1,405	1,084	698	321	15.9069	0.295330
26163545900	1	1,637	2,530	1,018	681	592	239	11.4077	0.211797
26163546000	1	1,464	2,008	748	554	450	256	9.0612	0.168231
26163546100	1	1,626	2,645	1,021	687	646	291	11.9076	0.221079
26163546201	1	1,245	2,214	773	645	540	256	10.0003	0.185667
26163546301	1	720	1,334	408	345	366	215	5.9947	0.111298
26163546600	1	873	1,407	525	349	354	179	6.3234	0.117402
26163546700	1	1,299	2,091	718	481	529	363	9.3748	0.174054
26163547000	1	1,730	2,586	753	595	679	559	11.5849	0.215088
26163547100	1	2,183	3,311	1,220	828	769	494	14.8922	0.276491
26163547200	1	1,737	2,753	981	620	635	517	12.3459	0.229216
26163985100	4	140	164	94	35	23	12	0.7380	0.013702
26163985300	2	247	209	120	50	32	7	0.9430	0.017508